

# YIMIAO YU

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## RESEARCH INTERESTS

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Science of deep learning and training dynamics: what training leaves in the weights, and how it shapes later learning.

## EDUCATION

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### New York University, Center for Data Science

New York, NY

Master of Science in Data Science; GPA: 3.78/4.0

Sep 2025 – May 2027 (Expected)

- Coursework: Machine Learning (A; top score in class, 100.2/100), Probability & Statistics, Optimization & Computational Linear Algebra, Big Data; Fall 2026: Deep Learning, Inference & Representation, Capstone Project

### Zhejiang Ocean University

Zhoushan, China

B.S. in Data Science & Big Data Technology; GPA: 3.69/5.0 ( $\approx 86.9/100$ )

Sep 2021 – Jun 2025

- Coursework: Algorithm Design & Analysis, Machine Learning, Data Visualization, Applied Multivariate Statistics

## PUBLICATIONS

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Yimiao Yu, Florentin Guth. *Localizing Transfer Between Memorization Tasks*. NeurIPS 2026 Workshop on Neural Network Artifacts as a New Data Modality (poster). [arXiv:2610.00771](https://arxiv.org/abs/2610.00771) | [Code](#)

## RESEARCH EXPERIENCE

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### Research Assistant — Transfer Between Memorization Tasks

New York, NY

NYU Center for Data Science; Advisor: Florentin Guth

Jan 2026 – Present

- Question:** networks can memorize random labels, a task with nothing to generalize. Does that memorization still speed up later learning, and where in the weights is the speedup stored? First-author project (CIFAR-10, CNNs)
- Found **equivalent transfer**: after a short warm-up, each epoch spent memorizing one set of random labels saves  $\approx 1$  epoch on a different set, even on disjoint images
- Found **non-equivalent transfer**: early in pre-training, each epoch on pure random-pixel noise saves *more* than one downstream epoch, briefly beating training on the downstream task itself
- Showed transfer needs a **consistent task, not repeated data**: fresh noise inputs every epoch still transfer though pre-training accuracy stays at chance; fresh labels every epoch (conflicting targets) remove all transfer
- Localized** the transfer with ablations, norm rescaling, and SVD swaps: a **magnitude-driven** part in the last layer, fully explained by weight-norm growth, and a **structure-driven** part in the conv layers, partly carried by the weight covariance; what carries the rest is open

## HONORS & AWARDS

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Third Prize (2022), Chinese Mathematics Competitions (CMC), Zhejiang Province

Second Prize (2022) and Third Prize (2023), FLTRP-ETIC Cup English Public Speaking Contest, Zhejiang Provincial Round

Third-Class Scholarship, Zhejiang Ocean University (2021)

## INDUSTRY EXPERIENCE

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### Capgemini

China

Data Analysis Intern; Lead, Vehicle-Model Recognition Project

May 2025 – Aug 2025

- Built a four-tier rule-based matching pipeline (exact, substring, brand-stripped, keyword lexicon) with a blacklist and manual review, identifying 826 new-energy vehicle models among 2,334 records
- Built a fault-tolerant Selenium scraper for historical sales data (checkpoint/resume, batched restarts, retries, SQLite error log)

## TECHNICAL SKILLS

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**Programming:** Python (primary), SQL, Java, C    **ML/Data:** PyTorch, NumPy, Pandas, Scikit-learn, SciPy, Matplotlib

**Tools:** Git, Jupyter    **Languages:** Mandarin (native), English (fluent)